ACM TKMCE Website – Project Report

ACM TKMCE Student Chapter, TKM College of Engineering, Kollam

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1. Current Status of the Website

The ACM TKMCE website is a complete website and event management portal for the chapter. Students can sign in, register for events, apply for ACM membership and message the chapter. Organisers manage everything from an admin panel without touching the code. The site is hosted on Vercel and uses Firebase for login and the database.

- **Live:** The public website, Google sign-in, events, execom, gallery, contact form, membership application and the admin panel.
- **Completed, waiting to be deployed:** Team events, automatic random grouping, the messaging inbox, college-based eligibility (TKM only / Inter-College), event guidelines and code of conduct, WhatsApp group links, and several admin panel improvements.
- **Built, not yet switched on:** Online payments through Razorpay for event fees and membership fees.
- **In progress:** A new 3D animated home page, kept as a separate version until it is finalised.

2. Options Included for Users

2.1 Visitors (without signing in)

- **Home page:** Chapter introduction, announcements ticker, about section with chapter statistics, why join ACM, vision, mission and values, and a photo gallery.
- **Events page:** Browse all events with filters: All, Latest, Registration Open, Ongoing, Coming Soon, Registered, Closed and Past.
- **Event details:** Poster, date, venue, description, speakers, guidelines and code of conduct. Each event has its own shareable link.
- **Live status:** A countdown before an event starts and a "Live Now" badge while it is running.
- **Execom page:** The executive committee, team by team, with member details.
- **Contact page:** Chapter email, LinkedIn and Instagram links.
- The site works on mobile, tablet and desktop.

2.2 Students (after signing in)

- **Sign in with Google.**
- **Profile:** Name, phone number, college, department, year and gender. Phone numbers can be verified by OTP (optional). The email address is locked to the Google account.

- **Event registration:** A simple step-by-step form, filled in automatically from the profile. Registration respects the event capacity, and events are either TKM students only or open to all colleges (Inter-College).
- **Team events:** The team leader creates a team and adds teammates by email. Teammates who don't have an account yet are linked automatically when they first sign in.
- **Random grouping events:** Students register individually and are placed into teams automatically.
- **ACM member discount:** ACM members can enter their membership ID to get a discount on event fees. The ID is verified first.
- **ACM membership:** TKM students can apply for membership and track its status: payment status and membership ID.
- **Dashboard:** Registered events, team members, WhatsApp group links and membership status.
- **Messages:** Send a message to the chapter and receive replies in the inbox, with a notification for unread replies.

3. Functionalities for the Admin

There are two levels of access. A **full admin** can use every section. Other members can be given access to **specific sections only** (Events, Execom, Content, Membership, Messages or Registrations), so work can be shared without giving full control.

Section	What the admin can do
Overview	See overall statistics, upcoming events, storage usage, and alerts for payments that need checking.
Events	Create, edit and delete events. Set date and time, venue, fee, maximum registrations, registration opening and closing dates, poster, guidelines, code of conduct and WhatsApp link. Mark an event as TKM only or Inter-College, enable the ACM member discount, and set up group events with team sizes and random grouping.
Registrations	View all registrations event by event, with filters. See team details, download registrations as a CSV (Excel) file, and message a participant or a whole team. Admins can also add or remove team members, delete teams, and run grouping manually.
Membership	View membership applications, search and filter them, assign membership IDs, export the list, and message members.
Messages	Read and reply to messages from students, and track which messages are unread or handled.

Section	What the admin can do
Execom	Add, edit, reorder and remove teams and members, including roles, bios and photos.
Site Content	Edit the home page text, statistics, vision, mission and values, announcements, and the department list, all without changing code.
Gallery	Upload photos (cropped and compressed automatically), add captions and delete photos.
Payments	Add Razorpay payment accounts, set the membership fee, and choose which account receives each payment.
Access	Search members and give or remove section-wise admin access.

4. Tech Stack Used

Area	Technology
Frontend	React 18, Vite, React Router
Styling	Tailwind CSS, Lucide icons
Animations	Framer Motion, GSAP, Lenis (smooth scrolling)
3D and visual effects	Three.js, OGL (React Bits components)
Login	Firebase Authentication (Google sign-in, phone OTP)
Database	Cloud Firestore (with security rules)
Server functions	Vercel Serverless Functions (Node.js), Firebase Admin SDK
Payments	Razorpay
Hosting	Vercel
Testing	Playwright (automated browser testing)

The website runs on free plans: Firebase Spark and Vercel Hobby.

5. Work Pending

5.1 Payment Integration

- Add the Razorpay account keys and configure them on the server (Vercel).
- Tighten the security checks in the database rules and payment verification before payments go live.
- Add a "Pay Now" option for event registrations whose payment was not completed.

- Show payment status on the student dashboard, and let admins mark offline payments (cash/UPI) as paid.
- Test the full payment flow in Razorpay test mode, including payment confirmation through the webhook.
- Complete Razorpay KYC and switch from test keys to live keys.

5.2 UI Improvements

- Finalise the new 3D animated home page and add it to the main website.
- Improve speed and smoothness on mobile phones: lighter animations and faster loading.
- Improve accessibility: keyboard navigation in pop-ups and full support for the reduced-motion setting.
- Add the Testimonials section.
- Add LinkedIn links for execom members.
- Design a proper "Page Not Found" page.
- Improve the dashboard and inbox: live message updates and an option to cancel a registration.

5.3 Minor Fixes

- Stop registrations for events that have already ended.
- Prevent an event from going over its capacity when many students register at the same time.
- Correct placeholder text and typos in the default content (statistics, announcements, vision, mission and values).
- In the admin panel, show "Saved" only after a save has actually succeeded.
- Membership form: fill in the student's details automatically and validate the phone number.
- Correct labels and messages that don't match how the site works.
- Remove unused code and update the setup guide (README).

5.4 Deployment

- Deploy the completed features to the live website, update the Firebase database rules, and add the required server keys on Vercel.